MODELO ARX PRUEBA PRBS

IDENTIFICACION DE PROCESOS USANDO MINIMOS CUADRADOS

```
clear all
close all

Datos=xlsread('Datos_PRBS.xlsx');

t=Datos(:,1);
U=Datos(:,2);
Y=Datos(:,4);

subplot(2,1,1)
plot(t,U,'* k','MarkerSize',4)
ylabel('Entrada')
grid on
subplot(2,1,2)
plot(t,Y,'o b','MarkerSize',4)
ylabel('Salida')
xlabel('Tiempo')
hold on
grid on

% Seleccionando na=3, nb=3, d=1
s=length(Y);

X=[-[0;Y(1:s-1)],-[0;0;Y(1:s-2)],-[0;0;0;Y(1:s-3)],[0;U(1:s-1)],[0;0;U(1:s-2)],[0;0;0;U(1:s-3)

%Matriz de covarianzas
Q=X'*X
```

```
Q = 7×7
10^4 ×
    9.5025    9.0327    8.4148   -0.2275   -0.2315   -0.2212   -0.2026
    9.0327    9.5025    9.0327   -0.2089   -0.2275   -0.2315   -0.2212
    8.4148    9.0327    9.5025   -0.1891   -0.2089   -0.2275   -0.2315
   -0.2275   -0.2089   -0.1891    0.0060    0.0055    0.0050    0.0045
   -0.2315   -0.2275   -0.2089    0.0055    0.0060    0.0055    0.0050
   -0.2212   -0.2315   -0.2275    0.0050    0.0055    0.0060    0.0055
   -0.2026   -0.2212   -0.2315    0.0045    0.0050    0.0055    0.0060
```

```
%Vector de parámetros
Theta=inv(Q)*X'*Y
```

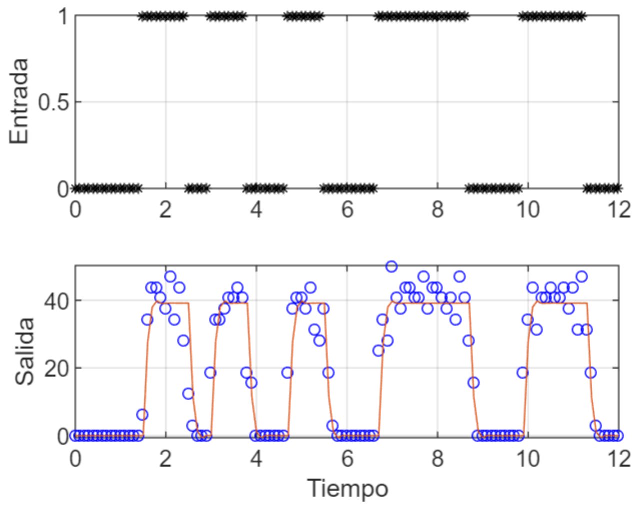
```
Theta = 7×1
   -0.1364
    0.0122
   -0.0156
   27.8529
    6.3541
    0.7226
```

```
   -1.2351
```

```
z=tf('z',0.1);

Gpm=((Theta(4)*z^3+Theta(5)*z^2+Theta(6)*z+Theta(7))/(z^3+Theta(1)*z^2+Theta(2)*z+Theta(3)))*(
```

```
Gpm =

   27.85 z^3 + 6.354 z^2 + 0.7226 z - 1.235
  ------------------------------------------
  z^4 - 0.1364 z^3 + 0.01225 z^2 - 0.01557 z

Sample time: 0.1 seconds
Discrete-time transfer function.
```

```
%step(Gpm,'r--',12)
grid on
Ym=lsim(Gpm, U, t);
plot(t,Ym)
```




```
%Ym=step(Gpm,12);

%Validación del modelo
Sy=sum((Y-mean(Y)).^2)
```

```
Sy = 4.4119e+04
```

```
Sr=sum((Y-Ym).^2)
```

```
Sr = 4.8740e+03
```

```
R2=(Sy-Sr)/Sy
```

```
R2 = 0.8895
```